

B.E. 4th Semester Theoretical Examination, 2022-2023

Subject: Environmental Science | **Course:** MCHU 402

Q1. Short Answers

- **Active vs Passive Solar:** Active uses mechanical/electrical devices (pumps). Passive uses building design/materials.
- **Open Dump Drawbacks:** Groundwater leachate, methane fires, disease vectors.
- **EIA Steps:** Screening, Scoping, Baseline Data, Impact Prediction, Mitigation, Public Hearing.
- **Threats to Biodiversity:** Habitat destruction, pollution, climate change, invasive species.

Q2. Disasters and Noise

- **Natural vs Man-Made:** Natural = Earthquakes. Man-Made = Bhopal Gas Tragedy.
- **Sound vs Noise:** Sound is sensory perception; Noise is unwanted, disruptive sound.

Q4. Ecosystem Dynamics

- **Decomposer Role:** Fungi/bacteria recycle dead organic matter into soil nutrients.
- **Biodiversity Levels:** Genetic, Species, Ecosystem.
- **Ecological Pyramid:** Graph showing biomass or energy flow at each trophic level.

Q5. EIA and Fuels

- **EIA Drawbacks (India):** Poor compliance, weak public participation, inadequate baseline data.
- **Fossil vs Bio-fuel:** Fossil (coal) adds ancient carbon. Bio-fuels (ethanol) from contemporary matter are carbon-neutral.

Q9. Bioremediation

- **In-situ vs Ex-situ:** In-situ treats soil where it lies. Ex-situ excavates it for treatment elsewhere.

- **Bio-magnification:** Heavy metals (mercury) accumulating in high concentrations in top predators.